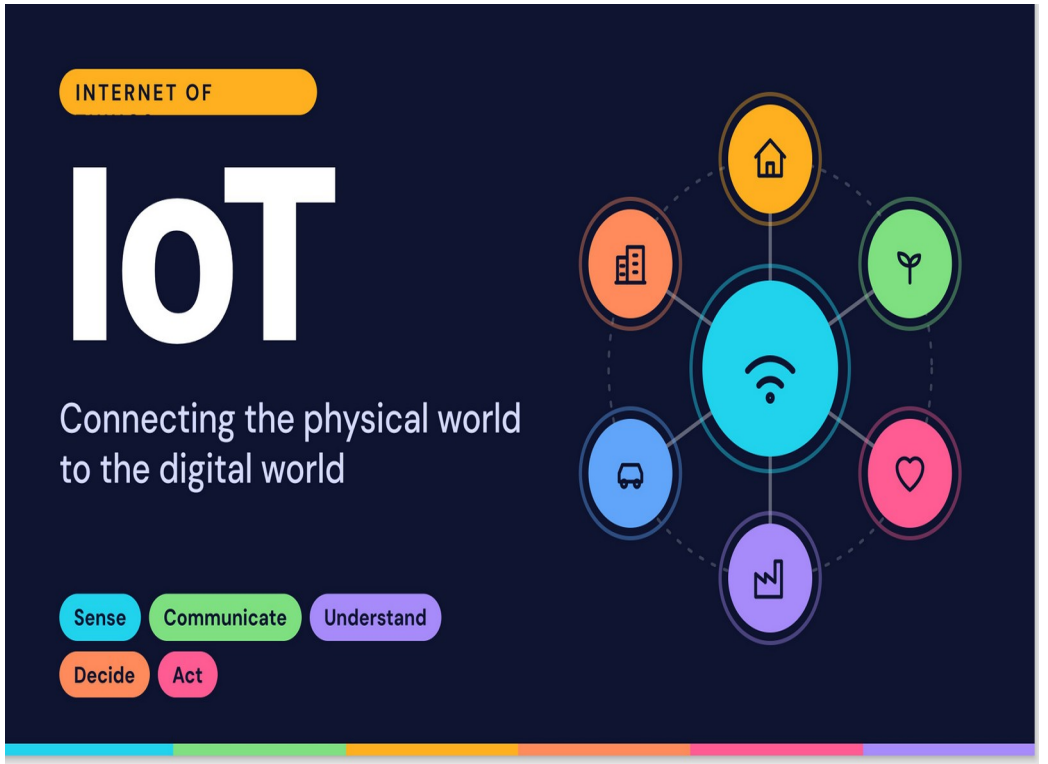

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VIZIANAGARAM
Knowledge Creates a Brighter Future
Knowledge for Tomorrow
Better Minds for a Brighter Society
Guidance for Today
Together Builds Governments
With Respect and Gratitude,
Dr. G Jaya Suma
Professor
JNTU GV VIZIANAGARAM

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IoT
FOUNDATIONS
From Devices to a Smarter World


INTERNET OF
IoT
Connecting the physical world to the digital world
Sense
Communicate
Understand
Decide
Act

What is IoT?

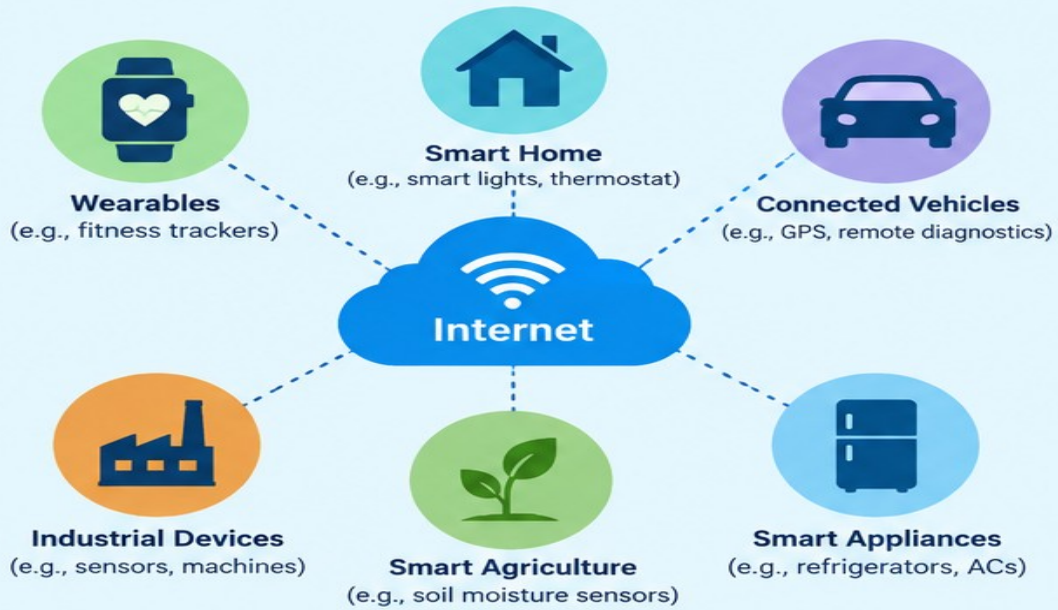
IoT (Internet of Things) refers to a network of physical devices, vehicles, appliances and other objects embedded with sensors, software and connectivity, enabling them to collect and exchange data over the internet.



Smart Devices.
Connected Data.
A Smarter World.

In Simple Words

IoT connects everyday devices to the internet, so they can sense, share and act on data without human intervention.



Sensors + Internet + Data = Smart Actions

Why is IoT Important?

IoT is transforming the way we live, work and interact with the world. It is important because:

Increases Efficiency
Automates processes, reduces manual work and saves time and resources.

Enables Better Decision-Making
Provides real-time data and insights for smarter, faster decisions.

Improves Safety and Security
Monitors environments, detects risks and prevents failures (e.g., smart homes, industrial safety).

Supports Sustainability
Optimizes resource usage and reduces waste and energy consumption.

Drives Innovation and New Services
Creates opportunities for smart cities, healthcare, transportation, agriculture and more.

Enhances Quality of Life
Provides convenience, personalization and better access to services.

IoT isn't just about connected devices — it's about a smarter, safer, more efficient and sustainable future.

2 Why does IoT matter so much today?

We are moving from a world of connected computers to a world of connected physical objects.

1 Everything is becoming connected

- Cars
- Watches
- Medical devices
- Machines
- Homes
- Agriculture

City infrastructure can all be connected.

Example: Smartwatch → heart rate → smartphone → alerts

2 IoT converts the physical world into data

Physical world → Sensors → Data → Analytics → Decision and action

Example: Temperature → sensor → digital data → AI analysis → high-temp alert → cooling system ON

3 Real-time monitoring

- Factory vibration
- Patient heart rate
- Soil moisture
- Traffic conditions
- Electricity consumption
- Water quality

IoT enables organizations to monitor systems 24/7.

4 Automation

Traditional system

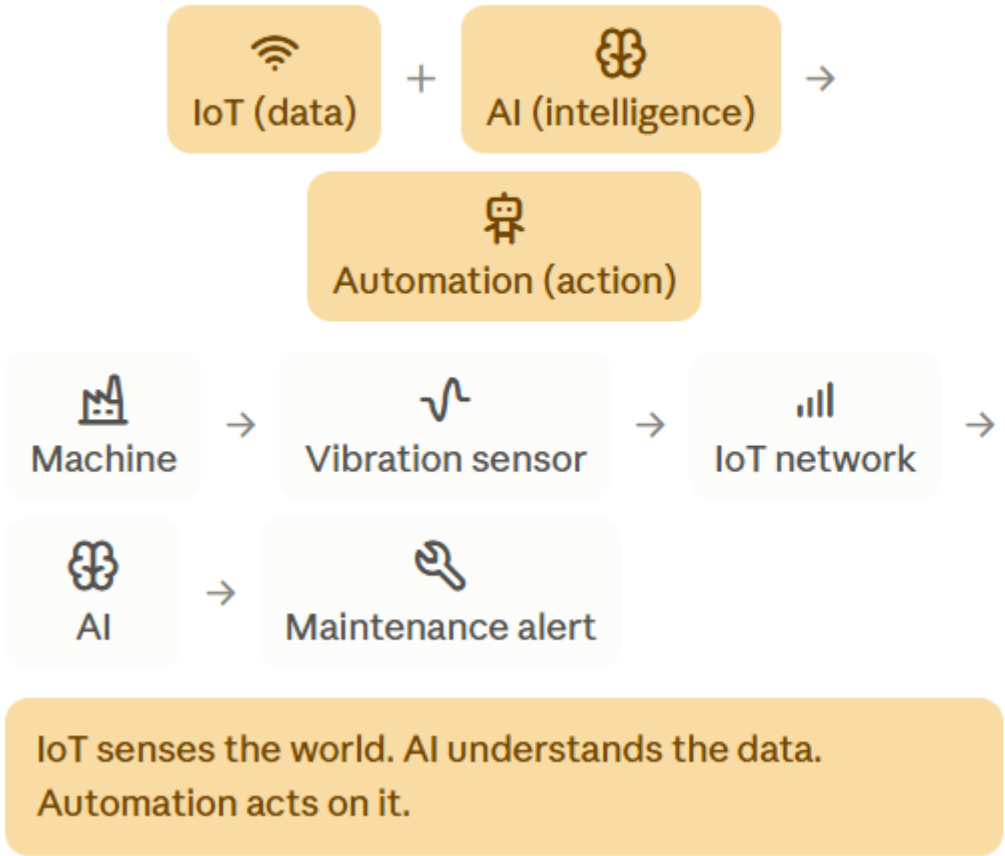
Observe → Decide → Act

IoT system

Sensor → Data → Intelligence → Automatic action

Reduces the need for continuous human intervention.

5 IoT + AI creates intelligent systems



IoT senses the world. AI understands the data. Automation acts on it.

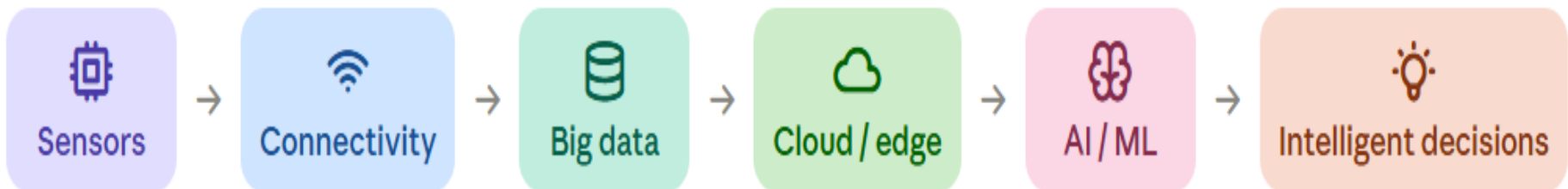
6 Improves efficiency and reduces cost

- Energy wastage
- Equipment failures
- Water consumption
- Production inefficiencies
- Resource utilization

Example: Smart lighting switches ON only when people are present.

5 IoT generates enormous amounts of data

Millions or billions of devices continuously produce data.



This is why IoT is closely connected with AI, cloud computing, edge computing, 5G, cybersecurity, and big data.

The evolution of connectivity

1. Past

Computers were connected

Machines were linked to share files, printers, and processing power. People stayed outside the network, and the computers waited for instructions from a human.

Example: an office network sharing files

2. Internet era

People and computers became connected

Email, websites, social media, and online shopping put people on the network. The network was global, but data still came mostly from humans typing, clicking, and uploading.

Example: a video call betw vo countries

3. IoT era

Physical things became connected

Watches, cars, machines, and farm equipment gained sensors and network links. They report temperature, location, or heart rate on their own, so the physical world starts producing digital data without anyone entering it.

Example: a smartwatch sending heart rate to a phone

↓

4. AI + IoT era

Connected things became intelligent

Connected devices produce huge amounts of raw data. AI analyzes it, spots patterns, and makes predictions, so devices move from just reporting to understanding.

Example: a vibration sensor helps predict a machine failure before it happens

5. Future

Things sense, communicate, understand, decide, and act autonomously

The whole loop runs with little or no human involvement. Systems notice a situation, work out the best response, and carry it out themselves.

Example: a cooling system switching on when a high temperature is detected

The autonomous loop in the Future stage

Sense
Sensors collect data from the physical world.

Communicate
Data travels over Wi-Fi, 5G, or low-power wireless.

Understand
Analytics and AI make sense of the data.

Decide
The system chooses the best response.

Act
Actuators carry it out, like cooling on or an alert sent.

One-line takeaway:
"IoT connects the physical world to the digital world, enabling us to sense, communicate, analyze, decide and act."

3 IoT ecosystem and future

IoT is transforming major sectors

Sector	IoT application
Smart home	Smart lights, security, appliances
Agriculture	Smart irrigation, soil monitoring
Healthcare	Wearables, remote patient monitoring
Industry	Predictive maintenance, automation
Transportation	Connected vehicles, fleet monitoring
Smart cities	Traffic, parking, waste management
Energy	Smart meters a id monitoring
Water	Leakage and quality monitoring

The big picture

From Connection to a Smarter World

Past	Computers were connected.
Internet era	People and computers became connected.
IoT era	Physical things became connected.
AI + IoT era	Connected things became intelligent.
Future	Sense → Communicate → Understand → Decide → Act

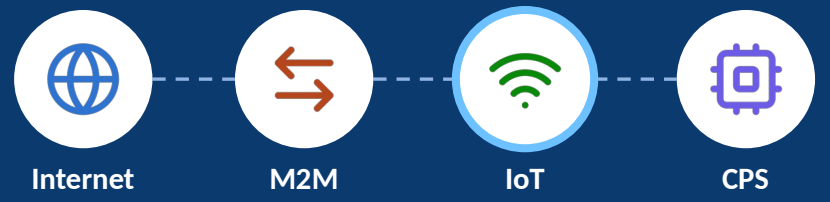
One-line takeaway

"IoT connects the physical world to the digital world, enabling us to sense, communicate, analyze, decide, and act."

Smarter devices | Connected world | A better future

IoT vs Related Concepts

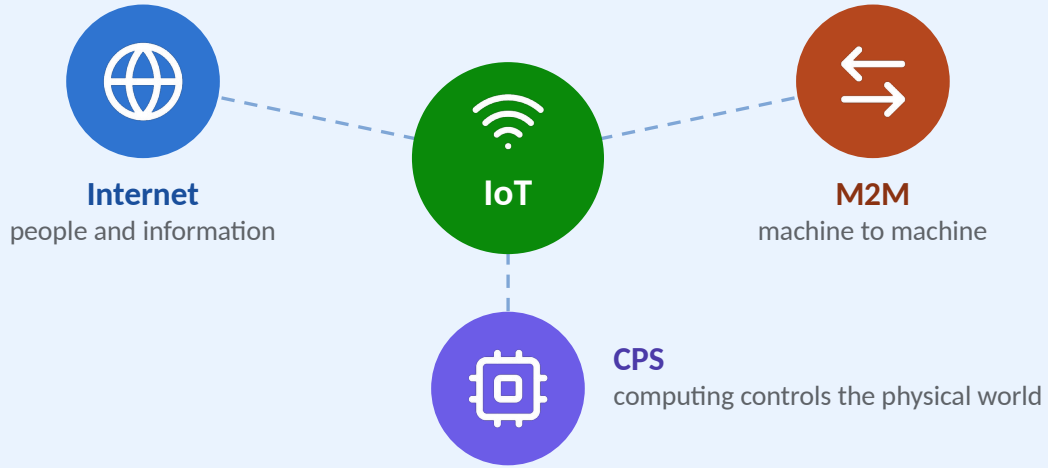
IoT is often confused with the Internet, M2M and Cyber-Physical Systems (CPS). They overlap and share technology, but each has a different focus and purpose.



Related. Overlapping. Not identical.

In Simple Words

The Internet connects people and information. M2M links machines directly. CPS uses computing to control physical processes. IoT connects everyday things to the Internet so they can sense, share and act on data.



How Is IoT Different?

IoT overlaps with each concept, but adds something more:

IoT vs Internet
The Internet is the network. IoT is the system of things, data and services that runs on it.

IoT vs M2M
M2M links machines directly. IoT connects many devices to shared cloud platforms and analytics.

IoT vs CPS
CPS focuses on real-time control of physical processes. IoT focuses on connectivity and data.

IoT overlaps with all three, but it is the one that links everyday things to the Internet, data and intelligence.

1 Why do these terms get confused?

They share the same technology, but they answer different questions.

1 They share the same building blocks

- Sensors
- Connectivity
- Software
- Data
- Actuators
- Computing

Each concept uses some mix of these ingredients.

Example: a smart thermostat has sensors, a network link, software and an actuator, so it touches all four ideas.

2 But each one asks a different question

Internet	How do we connect people and information?
M2M	How can machines exchange data on their own?
IoT	How can connected things share data and act smartly?
CPS	How can computing safely control physical processes?

Different questions, overlapping technology.

2

IoT vs Internet

The Internet connects people and information. IoT connects physical things.

Internet

the global network

CONNECTS

Computers, phones and people worldwide

DATA COMES FROM

Humans: typing, clicking, uploading

MAIN PURPOSE

Communication and access to information

EXAMPLE

A video call between two countries

vs

IoT

the network of things

CONNECTS

Physical things with sensors and software

DATA COMES FROM

Sensors and devices, automatically

MAIN PURPOSE

Monitor, analyse and act on the physical world

EXAMPLE

A smart thermostat adjusting room temperature

Key idea: the Internet is the network; IoT is a system of things, data and services that runs on it.

2

How the Internet and IoT work together

IoT needs connectivity, and the Internet gives it global reach.



Only the blue step is the Internet. The rest of the chain is what makes it IoT.

The Internet enables IoT

It gives devices global reach, so things can be monitored and controlled from anywhere.

Example: checking soil moisture on your phone from the city

IoT goes beyond the Internet

It adds sensors, actuators, analytics and automation on top of connectivity.

Example: a smart farm that also opens valves automatically

Note: not every IoT system needs the public Internet. Some run on local or private networks, such as a home hub linking Zigbee devices.

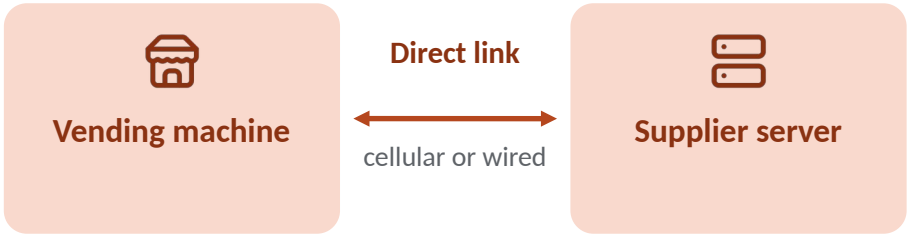
3

IoT vs M2M

M2M links machines directly. IoT connects many devices through the Internet and the cloud.

M2M

Machine-to-Machine



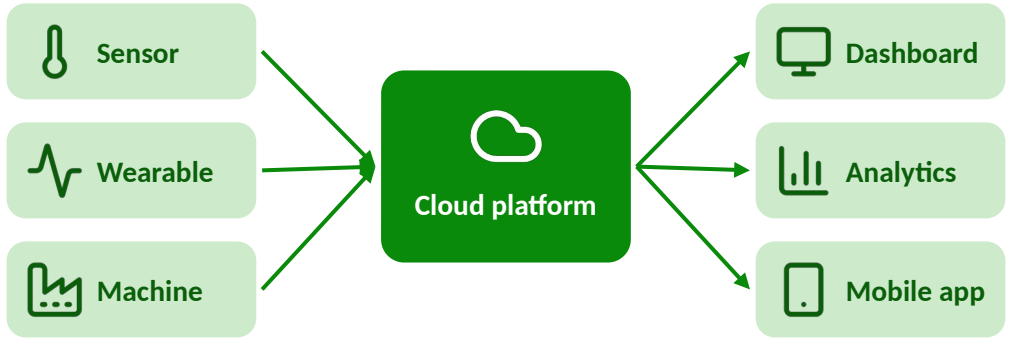
- Machines exchange data directly, no human needed
- Usually point-to-point and single-purpose
- Often closed or proprietary systems

Example: a vending machine reports low stock to a supplier's server

vs

IoT

Internet of Things



- Devices connect via the Internet to shared platforms
- Many devices can serve many applications
- Data feeds analytics, dashboards and AI

Example: a smart building's lights, HVAC and sensors feed one cloud platform

3

IoT vs M2M: side by side

M2M is often described as a building block, or predecessor, of IoT.

	M2M	IoT
Communication	Direct, point-to-point	Many-to-many via the Internet and cloud
Network	Cellular, wired or proprietary links	IP-based: Wi-Fi, cellular, LPWAN and more
Scope	Closed, single-purpose solutions	Open ecosystems, many applications
Data use	Basic monitoring and control	Analytics, AI and smart decisions
Scale	Small, fixed deployments	Highly scalable, up to millions of devices
Human role	Little or none	Users interact through apps and dashboards

Key idea: IoT takes the machine-to-machine link and adds Internet-scale integration, cloud and intelligence.

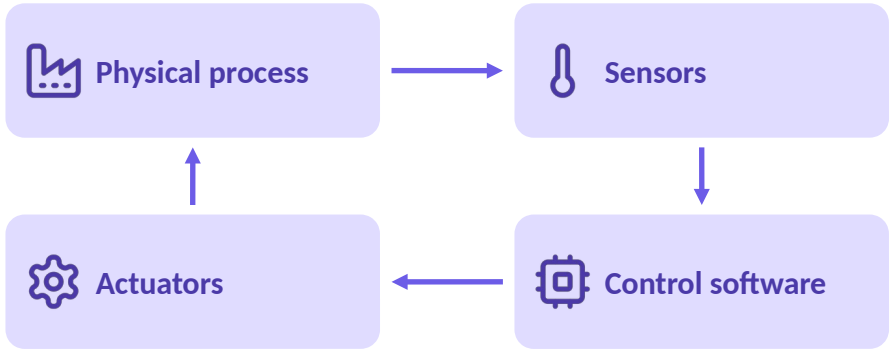
4

IoT vs Cyber-Physical Systems (CPS)

CPS tightly couples computing with physical processes. IoT connects things to share data and services.

CPS

Cyber-Physical Systems



A real-time feedback loop

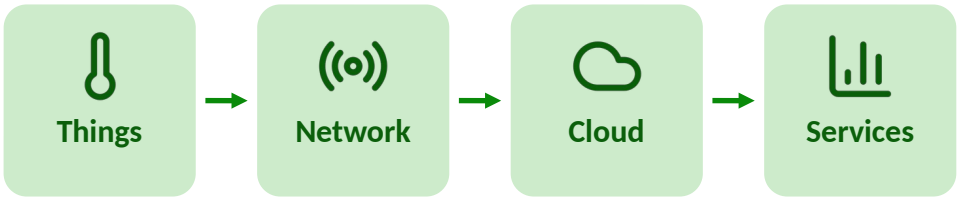
- Computers monitor and control physical processes
- Focus on real-time control, safety, reliability
- Need not use the public Internet

Example: a pacemaker senses heart rhythm and corrects it in real time

vs

IoT

Internet of Things



Data flows from things to analytics and services

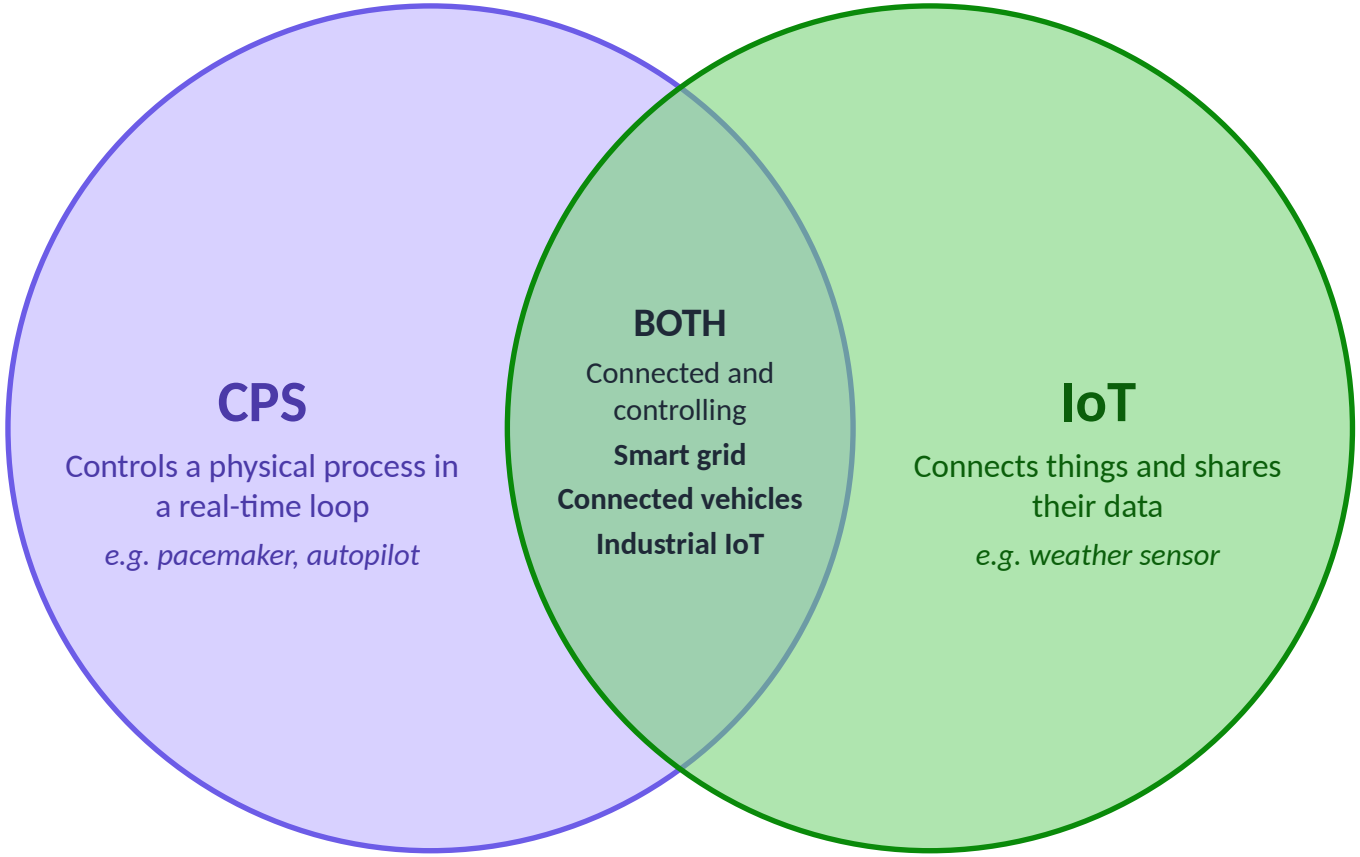
- Things sense, connect and share data over networks
- Focus on connectivity, data and services
- Some systems only monitor and never control

Example: a weather sensor sends readings to a cloud dashboard

4

IoT vs CPS: where they overlap

Many real systems are both. What matters is whether they connect, control, or do both.



A quick test

1. Is there a control loop?
Sensors feed computing, which drives actuators that change the physical process. If yes, it behaves like a CPS.

2. Is it networked?
Devices share data with the cloud, other systems or users. If yes, it behaves like IoT.

Yes to both?
Then the system is both IoT and CPS, for example a smart grid balancing supply and demand.

Comparison at a glance

	Internet	M2M	IoT	CPS
Connects	People, computers and information	Machines with other machines	Everyday things to the Internet	Computing with physical processes
Data comes from	Humans: typing, clicking, uploading	Machines, automatically	Sensors and devices, automatically	Sensors in the physical process
Main focus	Communication and information access	Automated data exchange between machines	Data, analytics and smart services	Real-time control, safety, reliability
Network	Global, IP-based	Point-to-point: cellular or wired	Internet-based and cloud-connected	Varies: embedded buses to the Internet
Example	A video call	Vending machine to supplier server	Smart thermostat or smart farm	Pacemaker or aircraft autopilot

The big picture

Internet	Connected people and information.
M2M	Connected machines, directly.
IoT	Connected everyday things, using the Internet, data and intelligence.
CPS	Computing tightly tied to physical processes for real-time control.
Remember	Internet = network · M2M = direct link · IoT = things + data · CPS = control

One-line takeaway

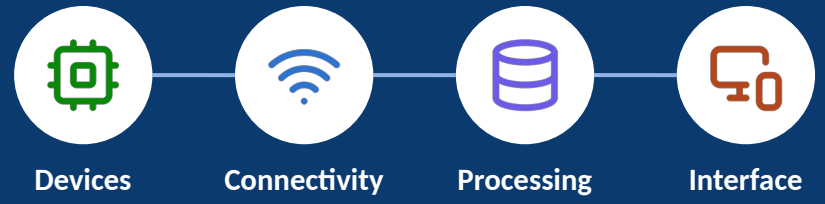
“The Internet connects people, M2M connects machines, CPS controls physical processes, and IoT connects everyday things to data and intelligence.”

Connected people | Connected machines | Connected things | Controlled processes

Up next: Building Blocks of IoT

Building Blocks of IoT

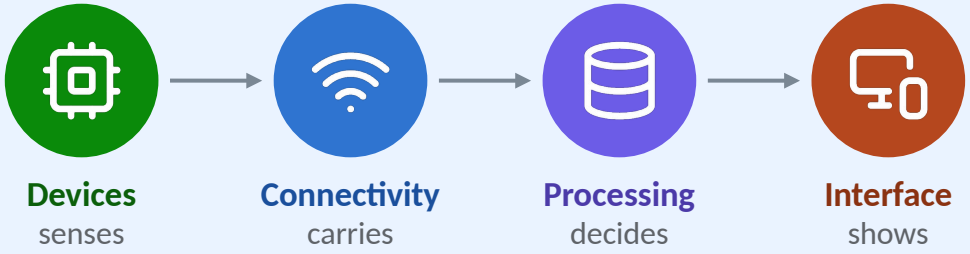
Every IoT system, from a smart bulb to a smart factory, is assembled from the same four components and powered by five enabling technologies.



Four blocks. One working system.

In Simple Words

An IoT system is a chain of four jobs. Devices sense the world, connectivity carries the data, data processing turns it into insight, and the user interface lets people see and act.



Feedback: commands flow back down to the devices.

Two Ways to Study Them

We look at the same system through two lenses:

Components
What the system is made of: devices, connectivity, data processing and user interface.

Enabling technologies
What makes each block possible today: sensors, connectivity, cloud, big data and AI.

Why it matters
With these blocks in mind you can read any IoT architecture, from a smart bulb to a smart city.

Four building blocks, five enabling technologies: one connected system.

1 Why break IoT into building blocks?

Every IoT system, big or small, does the same four jobs.

1 Think of IoT as a living body

Devices	**Senses and muscles** See, hear, touch and move.
Connectivity	**Nerves** Carry signals to and from the brain.
Processing	**Brain** Makes sense of signals and decides.
Interface	**Voice and face** How we see and talk to the system.

2 Two lenses on the same system

Part A: Components

What is an IoT system made of?

- Devices
- Connectivity
- Data processing
- User interface

Part B: Enabling technologies

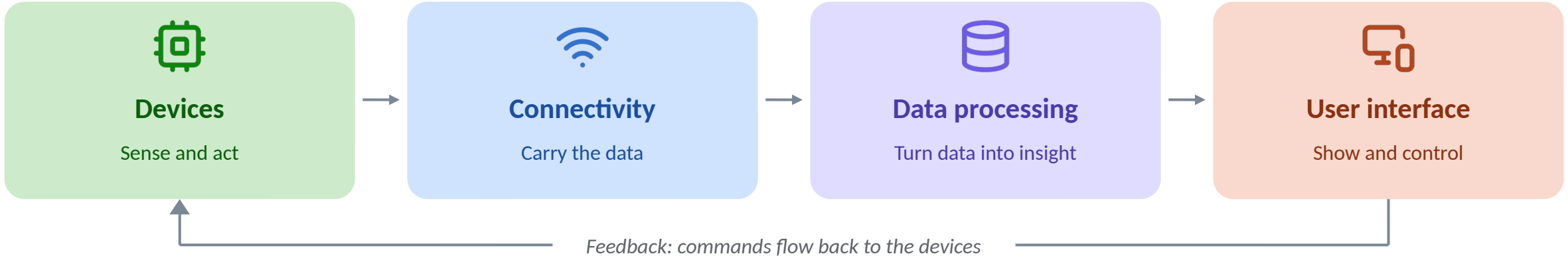
What makes each block possible today?

- Sensors
- Connectivity
- Cloud
- Big data
- AI

1

The four components of an IoT system

Data flows up from the physical world, and commands flow back down.



Follow the data: smart irrigation on a farm

1 Sense

A soil-moisture sensor measures the field: 28%.

2 Carry

A LoRaWAN radio sends the reading to a gateway, which forwards it online.

3 Decide

The cloud compares 28% with a target of 40% and flags the field as too dry.

4 Show and act

The farmer's app shows an alert. One tap, or an automatic rule, opens the valve.

Note: Small systems can merge blocks. A smart bulb and its phone app may talk directly, but all four jobs still happen.

2

Devices: the things that touch the physical world

Sensors read the world, actuators change it, and a small computer runs them.

Sensors

the input: they sense

JOB
Measure something in the physical world and turn it into an electrical signal.

EXAMPLES
Temperature, humidity, motion, light, soil moisture, GPS

SENDS
Readings to the microcontroller

Microcontroller

the brain of the device

JOB
Reads the sensors, runs the device program and prepares data to send.

EXAMPLES
ESP32, Arduino, NodeMCU

OFTEN INCLUDES
A radio such as Wi-Fi or Bluetooth to connect

Actuators

the output: they act

JOB
Take a command and make something happen in the physical world.

EXAMPLES
Relay, motor, valve, buzzer, smart lock, LED

RECEIVES
Commands from the microcontroller or the cloud

Key idea: devices are the only block that touches the physical world. Everything else works on data.

2

Connectivity: how the data travels

The right link depends on distance, power and how much data you send.

Short range

A few cm to about 100 m

Bluetooth LE | Zigbee | Wi-Fi | NFC

BEST FOR
Homes, wearables and smart buildings

TRADE-OFF
Limited range, so it usually needs a gateway to reach the Internet.

Long range, low power

Kilometres

LoRaWAN | NB-IoT | LTE-M

BEST FOR
Farms, meters and remote sensors on small batteries

TRADE-OFF
Low data rate: small messages, sent occasionally.

Long range, high speed

Wide-area network coverage

4G | 5G

BEST FOR
Vehicles, cameras and devices that send a lot of data

TRADE-OFF
Higher power use and data plan costs.

Shorter range → Longer range

Gateways bridge the gap. A gateway collects data from local links such as Bluetooth and Zigbee and forwards it to the Internet.

2

Data processing: from raw readings to decisions

Somewhere between the sensor and the screen, data must be stored, analysed and acted on.

Collect → **Store** → **Analyse** → **Act**

Readings arrive from many devices.

Kept as time-stamped records.

Rules and AI find patterns and problems.

Send alerts, update dashboards, trigger actuators.

Near the device (edge)

on the device or a local gateway

- Fast response: no round trip to the cloud
- Keeps working when the connection drops
- Sends only useful data, saving bandwidth

Example: a camera detects motion on the spot

+

In the cloud

on powerful remote servers

- Huge storage and computing power on demand
- Combines data from many devices and sites
- Trains and runs AI models on long histories

Example: the cloud spots pump vibration rising for weeks

Key idea: most real systems use both: the edge for speed, the cloud for scale.

3

Enabling technologies: why IoT works now

Five advances made connected devices cheap, reliable and useful at scale.



Sensors

WHAT CHANGED

Tiny, cheap, accurate and low-power sensors can now go almost anywhere.

EXAMPLE

A phone accelerometer or a soil probe



Connectivity

WHAT CHANGED

Wi-Fi, Bluetooth LE, LPWAN and 5G give devices low-power links at many ranges.

EXAMPLE

A LoRaWAN water meter



Cloud

WHAT CHANGED

On-demand storage and computing that grows with the number of devices.

EXAMPLE

A platform storing readings from thousands of sensors



Big data

WHAT CHANGED

Tools that store and process fast, never-ending streams of readings.

EXAMPLE

Years of vibration logs from every machine in a plant



AI

WHAT CHANGED

Machine learning finds patterns in data and turns them into predictions.

EXAMPLE

Predicting a bearing failure before it happens



Key idea: no single technology is IoT. Each one removes a barrier, and together they make connected things practical.

3

Which technology powers which block?

Each enabling technology strengthens one or more components.

Enabling technology	Devices	Connectivity	Data processing	User interface
Sensors	● Measure the physical world			
Connectivity	○ Radio chip inside the device	● Wi-Fi, BLE, LPWAN, 5G links		
Cloud			● Storage and computing on demand	○ Hosts apps and dashboards
Big data			● Handles huge, fast streams	
AI	○ Edge AI on the device		● Finds patterns and predicts	○ Voice assistants

● Main role ○ Also helps

Comparison at a glance

	Devices	Connectivity	Data processing	User interface
Main job	Sense the world and act on it	Carry data between devices, gateways and the cloud	Store, analyse and turn data into decisions	Show results and let people control the system
Typical tools	Sensors, actuators, microcontrollers	Wi-Fi, Bluetooth LE, Zigbee, LoRaWAN, NB-IoT, 5G	Edge gateways, cloud platforms, analytics, AI models	Dashboards, mobile apps, voice assistants, alerts
Example	A soil-moisture sensor and a valve	A LoRaWAN link from field to gateway	The cloud flags a field as too dry	The farmer's app shows 28% and opens the valve
If it is missing	No data comes in and nothing happens	Data stays trapped on the device	Only raw numbers, no insight	Nobody can see or control the system

The big picture

Devices	Sense the physical world and act on it.
Connectivity	Carry data between things, gateways and the cloud.
Data processing	Store, analyse and turn data into decisions.
User interface	Show results and let people monitor and control.
Remember	Devices sense · Connectivity carries · Processing decides · Interface shows

One-line takeaway

“Devices sense, connectivity carries, data processing decides, and the user interface shows, all made possible by sensors, connectivity, cloud, big data and AI.”

Devices sense | Connectivity carries | Processing decides | Interface shows

Up next: IoT Architecture

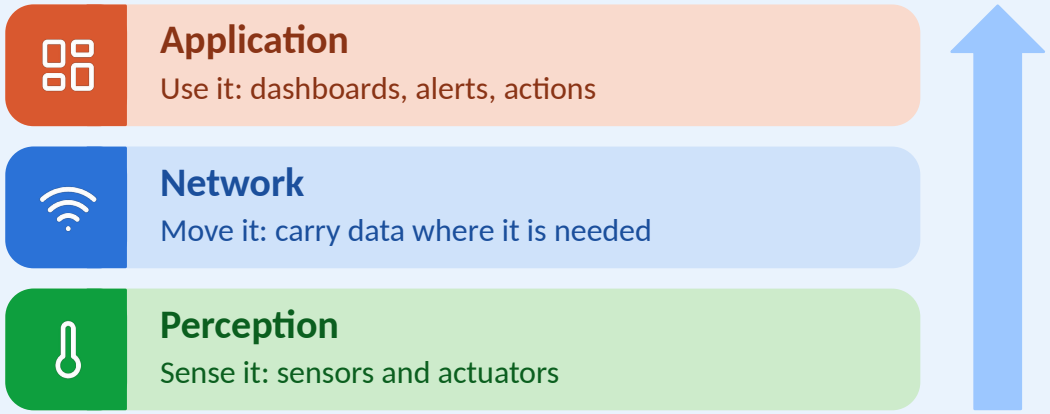
IoT Reference Architecture

A layered way of describing an IoT system, so heterogeneous devices, sprawling networks and diverse applications can work together as one organized whole.

Sense it. Move it. Use it.

In Simple Words

An IoT system is a stack of layers. Each layer has one job, and data flows up from the physical world to the person who uses it.



Perception + Network + Application = a working system

Why do we need an architecture?

Real IoT systems mix devices, networks and applications. A layered model keeps that manageable:

- **Tames complexity**
  Splits a big system into layers, one clear job each.
- **Handles heterogeneity**
  Many devices, networks and apps fit the same frame.
- **Works at any scale**
  From one smart apartment to a national power grid.
- **Gives a common map**
  Any IoT system can be mapped onto the same layers.
- **Shows where processing lives**
  Device, gateway or cloud becomes a design choice.

Architecture is the skeleton of every IoT system: learn it once, use it everywhere.

1 Two ways to describe an IoT system

Two reference architectures are in common use: a simple one and a more precise one.

3 Three-layer model

The simple view

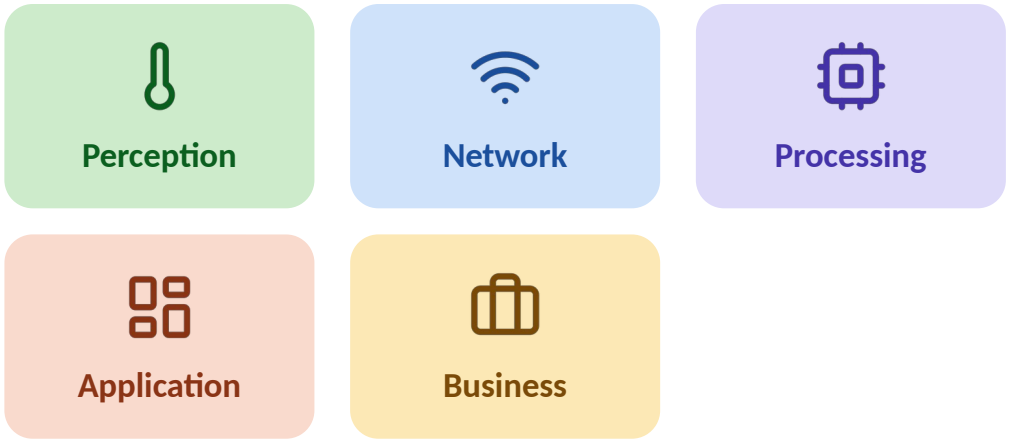


Sense it → Move it → Use it

Easy to hold in your head. It captures the core idea without unnecessary complexity.

5 Five-layer model

The more precise view



Same bottom two layers. The old Application layer is split into Processing, Application and Business.

Same system, two levels of detail: three layers show the shape of the data flow; five layers show what “use” really breaks down into.

The three-layer model

Data flows up the stack, from the physical world to the end user.

1. Perception

Sensors and actuators perceive the environment

Sensors perceive the surroundings, actuators act on them. Raw data collection happens here.

Example: a soil-moisture sensor buried at root depth

2. Network

Data is carried to where it needs to go

Carries the collected data onward using communication technologies and gateways (see Unit II).

Example: an ESP32 sending readings over Wi-Fi

3. Application

The service is delivered to the end user

A dashboard, an alert or an automated action: where collected data is finally put to use.

Example: a mobile app showing live soil moisture

Perception, Network, Application. Sense it, move it, use it.

3 Why a five-layer model?

In practice, the third layer quietly does an enormous amount of hidden work.

Application layer

in the three-layer model

- Storage
- Analytics
- Business logic
- User interface

One overloaded box doing everything.

splits into

Processing

Stores, analyzes and pre-processes the incoming data. Also called the middleware layer.

Application

Delivers processed information to user-facing services, such as apps and dashboards.

Business

Turns collected data into high-level business decisions and judges whether the system is succeeding.

Perception and Network stay exactly the same. Only the top layer is pulled apart.

3

The five-layer model

The full stack, mapped onto the three-layer view.

5 · Business	Turns data into high-level decisions and judges whether the deployment is succeeding.	Application in the 3-layer model
4 · Application	Delivers processed information to end-user services such as apps and dashboards.	
3 · Processing	Stores, analyzes and pre-processes the incoming stream (edge and cloud).	
2 · Network	Transmits the collected data to wherever it is processed.	Network in the 3-layer model
1 · Perception	Sensors and actuators perceive, and act on, the physical world.	Perception in the 3-layer model

Data flows up: sense → move → process → apply → decide

Layer 1 • Perception

A closer look at the physical layer.



Perception Layer

Where raw data collection happens, at the literal, physical point of sensing.

Sensors + Actuators

Sheer diversity

A fifty-cent mass-produced temperature sensor can sit beside a several-hundred-dollar precision vibration sensor in one deployment.

Passive and active

Passive sensors simply measure and report. Actuators physically change something about the world.

Harsh constraints

Most of a system's physical limits live here: battery life, durability against weather and vibration, size and unit cost.

Exposed to the world

These are the devices out in a field, worn on a wrist, or bolted onto a running industrial machine.

The system’s window onto the physical world: diverse, exposed and tightly constrained.

Layer 2 · Network

Moves data reliably from where it is sensed to where it is processed.



What makes this layer hard

Unreliable links

Data often travels over inherently unreliable wireless links.

Limited power

Devices frequently run on extremely limited battery power.

Huge distances

Data sometimes has to cross genuinely large geographic areas.

One of the hardest engineering problems in IoT, which is why Unit II is dedicated to it.

Layers 4 & 5 · Application and Business

Application Layer

Where a person, or another automated system, gets real value from everything below.

- **A clear, uncluttered dashboard**
- **A perfectly timed, relevant alert**
- **A simple, reliable automated action**

How successful a deployment feels to users depends on how thoughtfully this layer is designed, even when every layer beneath it is technically flawless.

Business Layer

Looks at the whole system and its business model, not at any single sensor reading.

- **Analyzes collected data to inform high-level business decisions**
- **Manages overall system operation strategically**
- **Evaluates whether a deployment is succeeding or failing**

Easy to overlook from a purely technical background, but it absolutely matters in real deployments.

5 Where does the processing happen?

A central theme of both models: edge, fog or cloud.

Closer to the device — Farther from the device

Edge computing

WHERE

On, or very near, the device that generated the data.

WHAT IT DOES

Runs a lightweight check, such as anomaly detection, on the microcontroller itself.

IMPACT

Lowest latency and less network bandwidth used.

Fog computing

WHERE

An intermediate tier on a local gateway or modest on-premises server.

WHAT IT DOES

Aggregates data from many edge devices, then sends a smaller, filtered volume onward.

IMPACT

A local “middle manager” that spares the cloud the raw firehose.

Cloud-centric

WHERE

A distant, centralized data center.

WHAT IT DOES

Every raw reading is shipped wholesale for central processing.

IMPACT

Stops scaling gracefully with thousands or millions of devices.

One of the most important design decisions in practical IoT. We go much deeper in Unit III.

6 Worked example: mapping a real system

Smart irrigation on a working farm, placed onto the five layers.

Smart irrigation deployment

Layer	What it looks like on the farm
Perception	Soil-moisture sensors buried at root depth
Network	An ESP32 microcontroller sending readings over local Wi-Fi
Processing	A cloud service that filters and stores every incoming reading
Application	A mobile app showing the farm manager live soil moisture
Business	Monthly report review: should irrigation expand to more acreage?

Once this clicks, you can run the same five-second mental exercise on any IoT system you meet.

The big picture

3 layers	Perception, Network, Application: sense it, move it, use it.
5 layers	Application splits into Processing, Application and Business.
Any system	Every real IoT system maps cleanly onto these layers.
Edge · Fog · Cloud	Where processing happens is a key design decision.
Up next	The devices that live in the Perception Layer.

One-line takeaway

“IoT is organized in layers: sense it, move it, process it, use it, and decide what it means for the business.”

Perception | Network | Processing | Application | Business

IoT Devices Explained

The real hardware of the Perception Layer: the sensors, actuators and small computers you will personally wire up and program in the Unit III labs.

Sense it. Compute it. Act on it.

In Simple Words

Every IoT device does some mix of three jobs: it senses the world, thinks locally, and acts back on it.

Sensor
The eyes and ears of the system

Microcontroller
The local brain, right beside the hardware

Actuator
The hands of the system

Sense + Compute + Act = an IoT device

The five device families

This topic tours each family in turn, from a single sensor to a finished smart product:

Sensors
Turn physical quantities into readable signals.

Actuators
Turn digital commands into physical action.

Microcontrollers
Provide local, on-site computation.

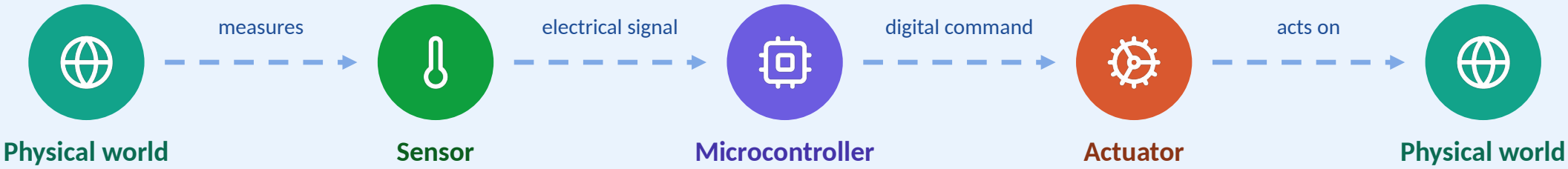
RFID, NFC and GPS
Give a device its identity and location.

Smart devices
Finished products that bundle it all together.

Small hardware details carry very real consequences for how a system performs in the field.

1 Sensors and actuators: a two-way interface

Together they give an IoT system its full connection to the physical world.



Sensor

The system's eyes and ears

Converts a physical quantity in the real world into an electrical signal a computer can read.

- Temperature
- Light level
- Motion
- Gas

Actuator

The system's hands

Does the exact reverse: converts a digital command from software into real physical action.

- Open a valve
- Switch a relay
- Spin a motor

Sensors let the system perceive its surroundings; actuators let it act on those same surroundings.

2 Sensors: a guided tour

Four sensors you will see constantly in the course and its lab series.

Common IoT sensors

Sensor	Measures	How it works
DHT11 / DHT22	**Temperature and humidity**	Extremely common and inexpensive: in nearly every beginner IoT project ever built.
LDR	**Ambient light intensity**	A light-dependent resistor: its resistance changes with how much light hits it.
HC-SR04	**Distance (ultrasonic)**	Sends a pulse of sound and times the returning echo, like a bat navigating in the dark.
PIR sensor	**Motion**	Passive infrared: senses changes in infrared (body heat) moving across its view.

You will wire these up yourself in the Unit III labs.

3 Actuators: the system's hands

The reverse of a sensor: a digital command becomes physical change.

Relay

Physically switches a circuit on or off.

Example: switching a water pump on

Servo motor

Turns a command into controlled motion.

Example: opening a ventilation vent

LED

Changes its state to show what is going on.

Example: signaling a status

Sensor perceives → **Microcontroller decides** → **Actuator acts**

Sensors and actuators together give an IoT system its full two-way interface with the world.

Microcontrollers and embedded systems

Embedded system

Any computing system built directly into a device to perform a small, dedicated set of functions.

Microcontroller-based IoT node

A small, self-contained computer on a single chip

- **Processor core**
- **Working memory**
- **Input and output pins**

One chip, three jobs

Reads your sensors

Takes readings from the sensors connected to it.

Runs your control logic

Executes whatever logic the application needs.

Drives your actuators

Switches connected actuators in response.

All in one tight package.

A microcontroller-based IoT node is one especially common example of a broader embedded system.

Three platforms you will build with

Hands-on projects in Unit III.

	Arduino	ESP32	Raspberry Pi
WHAT IT IS	Simple, robust development boards	Adds built-in Wi-Fi and Bluetooth right on the chip	A genuine general-purpose computer running a full Linux OS
STANDOUT	Genuinely beginner-friendly	Probably the most popular platform for networked IoT	Considerably more powerful, but more power-hungry and complex
BEST FOR	Learning the basics	Networked IoT projects	Heavier, general-purpose jobs

Do not memorize the differences yet. We compare them feature by feature in Unit III.

5 Identity and location: RFID, NFC and GPS

Without an identity, a device cannot really be a “thing” on the network.



RFID

A reader wirelessly reads a unique ID from a tag

No line of sight needed. Passive tags have no power source: they draw it from the reader’s signal.

Example: asset tracking, secure access badges



NFC

RFID’s short-range cousin

Works over a few centimeters at most. That short range is a feature: it keeps contactless payment secure.

Example: contactless payment



GPS

A device reports its own precise location

Triangulates radio signals from a constellation of orbiting satellites.

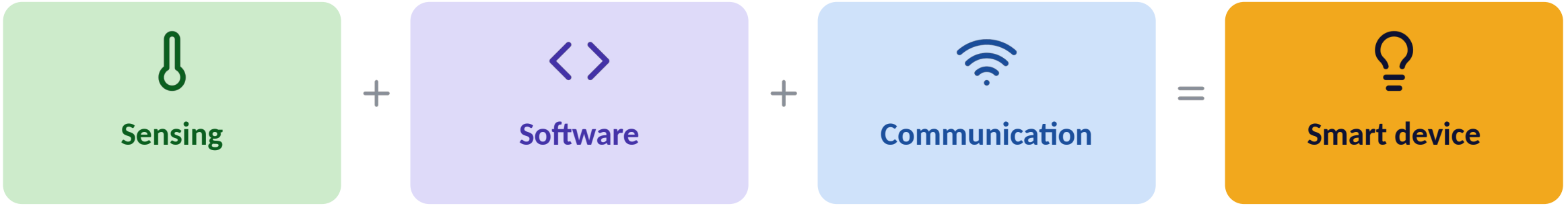
Example: fleet tracking, “your delivery is nearby”



Identification and location are prerequisites for treating an object as something addressable on a network.

6 Smart devices: everything in one product

The catch-all name for a finished product with sensing, software and connectivity built in.



Examples

Smart thermostat

Senses temperature, runs control logic and connects to an app.

Smart refrigerator

A household appliance with sensing, computing and connectivity built in.

Smart streetlight

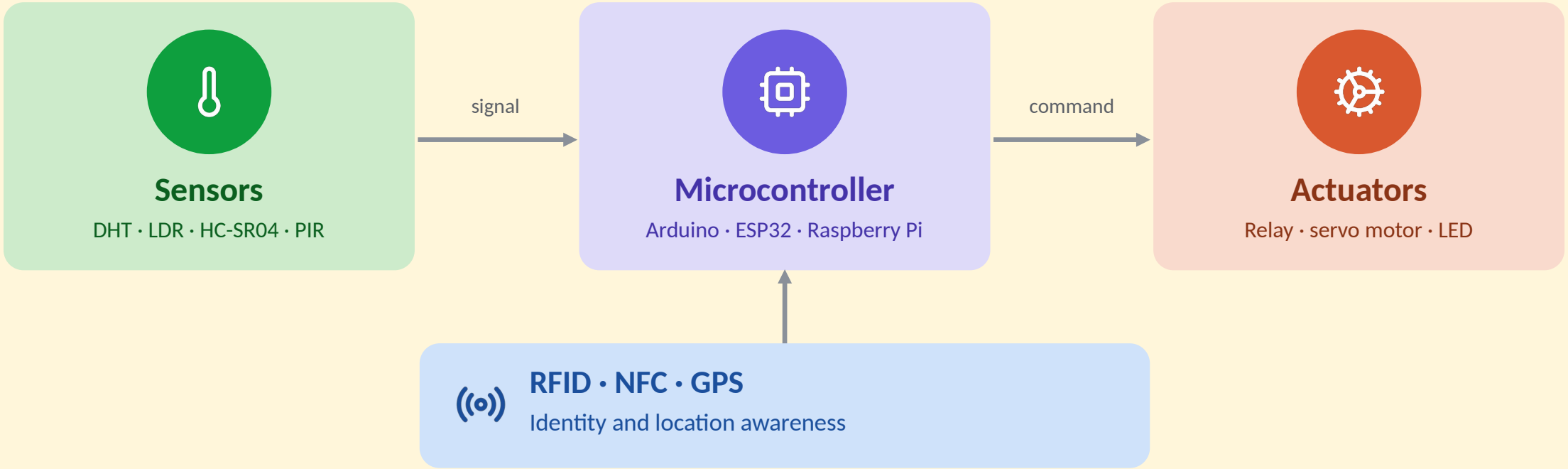
Stands on a city corner and can dim automatically at night.

The same three-part combination from Topic 2, wrapped into one consumer or industrial product.

How the devices fit together

One product, the same building blocks.

A smart device



Whatever the product, it is the same small set of device types combined in different ways.

Worked example · Battery life

How long will one battery last?

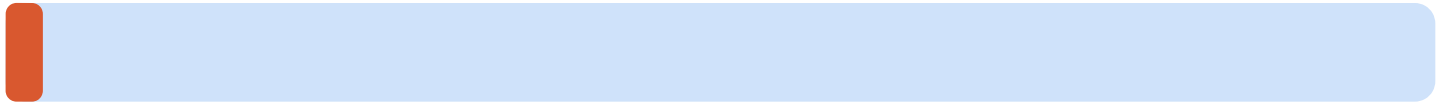
80 mA
while transmitting, for 2 s per reading

0.5 mA
while idle between readings

10 min
between readings (a 600 s cycle)

2,000 mAh
battery capacity

One 600-second cycle



2 s transmitting at 80 mA — 598 s idle at 0.5 mA

(transmit slice enlarged for visibility)

1. **Average current**
   (2 s × 80 mA + 598 s × 0.5 mA) ÷ 600 s
2. **Work it out**
   (160 + 299) ÷ 600 ≈ 0.765 mA
3. **Runtime**
   2,000 mAh ÷ 0.765 mA ≈ 2,600 hours

The node is idle for 598 of every 600 seconds.

The big picture

Sensors	Turn the physical world into usable digital data.
Actuators	Turn digital data back into real physical action.
Microcontrollers	Provide the local computational brains that tie it together.
RFID · NFC · GPS	Give devices their own identity and location awareness.
Smart devices	Finished products that bundle all of this into one package.

One-line takeaway

“Sensors perceive, actuators act, microcontrollers think, and small device-level details have very real consequences in the field.”

Sensors | Actuators | Microcontrollers | RFID · NFC · GPS | Smart devices

Up next: following one piece of data from sensor to dashboard.

The IoT Data Journey

The path one reading takes from a raw physical measurement to a meaningful decision or action, in five stages.

Measure. Move. Store. Analyze. Act.

In Simple Words

Think of one sensor reading as a traveler. It starts in the physical world and passes through five stages until a person, or a machine, acts on it.

1	**Acquire**	The sensor samples the world
2	**Transmit**	The reading moves onward
3	**Store**	Timestamped and persisted
4	**Analyze**	Insight is extracted
5	**Monitor**	Results reach people or machines

Why treat it as one pipeline?

A weakness at any one stage undermines everything downstream of it:

Sampling sets the ceiling
Analytics cannot recover what was never captured.

Dropped readings leave gaps
Unreliable transmission quietly corrupts the data.

Gaps mislead analytics
A time series with holes can point to wrong answers.

Clutter hides value
A confusing dashboard wastes all earlier engineering.

One connected whole
Design the pipeline together, not as five boxes.

Every application in Unit V builds on this same five-stage backbone.

1 The five-stage pipeline

One reading, five stages, from the physical world to a real decision.

1 Acquisition

A sensor samples the physical world, producing a raw digital reading.

2 Transmission

The reading moves across the network, via a gateway or straight to the cloud.

3 Storage

Timestamped readings are kept in a time-series database.

4 Analytics

Stored data is processed for insight, from simple thresholds to machine learning.

5 Monitoring

Results reach a person on a dashboard, or an actuator, closing the loop.

Perception layer	Network layer	Processing layer	Processing layer	Application layer

Where each stage lives in the five-layer architecture from Topic 5

The same pipeline recurs, in specialized form, across every application in Unit V.

Stage 1 · Data acquisition

A sensor samples the physical world.

A physical quantity → Sampled at a chosen rate → One raw digital reading

Sample far too rarely

You risk missing important, fleeting events entirely.

The sweet spot

A real trade-off between data fidelity and power and bandwidth cost. Getting the balance right is a genuine engineering skill.

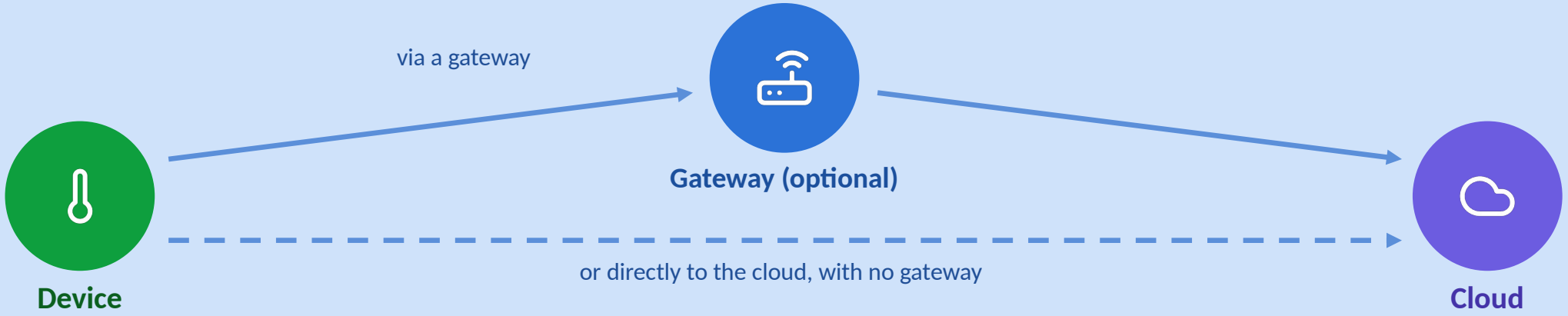
Sample far too often

You needlessly burn battery life and network bandwidth for very little extra benefit.

The sampling rate is a design decision in its own right, not an afterthought.

Stage 2 · Data transmission

The reading physically moves.



Network-layer technology

The reading crosses whichever technology was chosen for the deployment.

Gateway or direct

It heads onward to a gateway, or sometimes travels straight to the cloud.

The subject of Unit II

This entire stage is, essentially, the whole subject of Unit II.

If transmission silently drops readings, every stage after it inherits the gaps.

Stage 3 · Data storage

Readings are persisted for later use.

Timestamped readings

Timestamp	Sensor	Value
09:00:00	node-12	24.1 °C
09:05:00	node-12	24.3 °C
09:10:00	node-12	24.2 °C
09:15:00	node-12	24.6 °C

Illustrative readings

↓

A time-series database

Built for one recurring pattern

Huge volumes

Absolutely huge volumes of sequential, timestamped records.

Constantly written

New records keep arriving, written in real time.

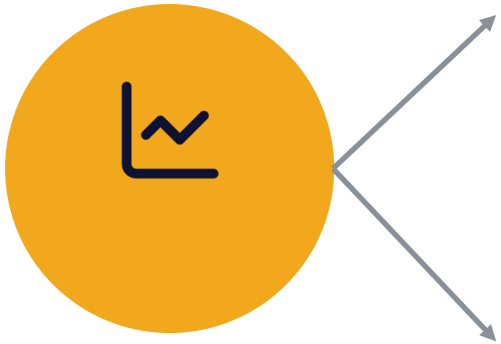
Occasionally queried

Read back now and then over a specific range of time.

Timestamped readings must be reliably persisted somewhere for later use.

Stage 5 · Real-time monitoring

Results finally reach someone or something.



Fully processed results

A person, through a dashboard

Results are presented to a human operator on a dashboard or mobile app.

Example: a farm manager checking live soil moisture

A machine, through an actuator

Results are fed back automatically to an actuator, closing the loop between sensing and acting.

Example: a soil sensor triggering automated irrigation

The loop closes

Physical world → Five-stage pipeline → Actuator → Physical world

The loop between sensing and acting that we first met in Topic 2.

Worked example · Data volume

A smart-city air-quality network.

500
air-quality sensors across a metropolitan area

200 bytes
in each reading

5 min
between readings, continuously

How much?
total data generated in one full day

1 **Readings per sensor per day**

$24 \times 60 = 1{,}440 \text{ min} \div 5 = 288 \text{ readings}$

2 **Data per sensor per day**

$288 \times 200 \text{ B} = 57{,}600 \text{ B} \approx 56 \text{ KB}$

3 **Whole deployment per day**

$500 \times 56.25 \text{ KB} \approx 28{,}000 \text{ KB} \approx 27.5 \text{ MB}$

About 27.5 MB a day sounds modest: less than one high-resolution smartphone photo.

2 Why every stage matters together

A weakness at any one stage undermines everything downstream of it.

1 · Acquisition → 2 · Transmission → 3 · Storage → 4 · Analytics → 5 · Monitoring

Stage 1

A poorly chosen sampling rate

↓

No clever analytics in stage 4 can recover information that was never captured.

Stage 2

Transmission that silently drops readings

↓

Stage 3 ends up with gaps in its time series that quietly mislead the analytics.

Stage 5

A cluttered, confusing dashboard

↓

All the careful engineering in the first four stages can still fail the person who needed to see it.

Treat the pipeline as one connected whole, not five boxes optimized in isolation.

3 The same backbone in Unit V

Two applications, one five-stage pipeline (illustrative sketches).

From sensor to action	Smart irrigation	Wearable patient monitor
Acquisition	Soil-moisture reading	Blood-oxygen reading
Transmission	ESP32 over local Wi-Fi	Wireless link to the hospital network
Storage	Timestamped readings in a time-series database	Timestamped readings, kept per patient
Analytics	Is the soil too dry?	Has blood oxygen dropped?
Monitoring	Dashboard, then automated irrigation	Urgent alert to the nurse

The full applications, with far more detail, arrive in Unit V.

The big picture

1 · Acquisition	A sensor samples the world at a chosen rate.
2 · Transmission	The reading travels via a gateway or straight to the cloud.
3 · Storage	Timestamped readings are persisted in a time-series database.
4 · Analytics	Stored data is analyzed, from thresholds to machine learning.
5 · Monitoring	Dashboards, or actuators, put the results to use.

One-line takeaway

“Every reading travels the same five stages, from acquisition to monitoring, and a weakness in any one undermines everything after it.”

Acquisition | Transmission | Storage | Analytics | Monitoring

Up next: a Unit I recap and a five-question self-check.

Unit I Recap & Self-Assessment

Pull the whole unit together in a rapid recap, then check your own understanding with five short questions.

Recap it. Test it. Move on.

Unit I in Five Questions

By now you can answer the big questions that frame the whole field:

1	**What is IoT?**	Objects that sense, share and act
2	**How is it different?**	From the Internet, M2M and CPS
3	**What is it made of?**	Sensors, actuators, gateways, cloud
4	**How is it organized?**	Layers, plus edge, fog and cloud
5	**How does data flow?**	Five stages, sensor to dashboard

How the self-check works

Five questions, one at a time, with a pause after each:

- **Read the question**
  One at a time
- **Pause for about five seconds**
  Resist the urge to peek
- **Answer on your own**
  Out loud, or in writing
- **Reveal the answer**
  Check it against yours
- **Revisit what was tricky**
  Every question maps to a topic

You will retain far more by testing yourself than by passively nodding along.

1 Recap: what IoT is, and is not

Topics 2 and 3.

Physical objects embedded with sensing, software and communication hardware, exchanging data over a network and very often acting back on the world.

It is clearly distinguished from three related ideas

The Internet

IoT focuses on physical objects that operate under severe power and processing constraints.

M2M

IoT uses shared, internet-based infrastructure rather than fixed, proprietary, point-to-point links.

Cyber-physical systems

IoT generally does not require the same tight, hard real-time control loop.

IoT is best seen as the broad, network-centric umbrella that connects physical devices at massive scale.

2 Recap: the building blocks

Topic 4: the same recurring parts in every IoT system.

Every IoT system is built from

- Sensors
- Actuators
- Microcontrollers
- Connectivity
- Gateways
- Cloud platforms

The same skeleton, whatever the domain.

Made possible by five enabling technologies

- Identification technologies
- Communication technologies
- Computation platforms
- Cloud and edge computing
- AI-driven analytics

Enabling technologies had to mature and become affordable before IoT at scale was possible.

3 Recap: the reference architecture

Topic 5: two views of the same layered system.

5 · Business	Business-level decisions; is the deployment succeeding?	Application in the 3-layer model
4 · Application	User-facing services: apps and dashboards.	
3 · Processing	Stores, analyzes and pre-processes the data.	
2 · Network	Transmits data to where it is processed.	Network in the 3-layer model
1 · Perception	Sensors and actuators at the point of sensing.	Perception in the 3-layer model

Edge and fog computing decide where processing happens: on the device, at the fog, or in the cloud.

4 Recap: the data journey

Topic 7: the backbone under every Unit V application.

1 Acquisition

A sensor samples at a chosen rate.

2 Transmission

The reading moves via a gateway or straight to the cloud.

3 Storage

Timestamped readings go into a time-series database.

4 Analytics

From simple thresholds to machine learning.

5 Monitoring

Dashboards, or actuators, put results to use.

Sampling rate is a design choice

Too rarely misses fleeting events; too often burns battery and bandwidth.

Data volume scales linearly

500 sensors made about 27.5 MB a day; 50,000 would make nearly 3 GB.

The exact same backbone every application in Unit V will build on.

Q1 Self-assessment: question 1 of 5

Tests: Related concepts (Topic 3)

What is the single biggest architectural difference between classic M2M communication and genuine IoT?

Pause the video

Try to answer on your own first. You retain far more by testing yourself.

about 5 seconds

5 4 3 2 1

The answer is on the next slide. No peeking!

Q1 Question 1: the answer

Tests: Related concepts (Topic 3)

What is the single biggest architectural difference between classic M2M communication and genuine IoT?

M2M relies on fixed, proprietary, point-to-point links to exactly one counterpart, while IoT uses shared, internet-based infrastructure and joins a much broader platform ecosystem.

Classic M2M

A fixed, proprietary, point-to-point link to one counterpart.

Genuine IoT

Shared, internet-based infrastructure and a broader platform ecosystem.

If this was tricky, revisit Topic 3: IoT versus related concepts.

Q2

Self-assessment: question 2 of 5

Tests: IoT architecture (Topic 5)

Name the three layers of the simple three-layer IoT architecture, in their correct order.

Pause the video

Try to answer on your own first. You retain far more by testing yourself.

about 5 seconds

5 4 3 2 1

The answer is on the next slide. No peeking!

Q2 Question 2: the answer

Tests: IoT architecture (Topic 5)

Name the three layers of the simple three-layer IoT architecture, in their correct order.

Perception, Network and Application.



If this was tricky, revisit Topic 5: IoT architecture.

Q3

Self-assessment: question 3 of 5

Tests: IoT architecture (Topic 5)

In the more detailed five-layer model, which layer from the original three gets split apart, and what are the three resulting layers called?

Pause the video

Try to answer on your own first. You retain far more by testing yourself.

about 5 seconds

5 4 3 2 1

The answer is on the next slide. No peeking!

Q3 Question 3: the answer

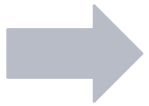
Tests: IoT architecture (Topic 5)

In the more detailed five-layer model, which layer from the original three gets split apart, and what are the three resulting layers called?

The Application Layer is split into the Processing Layer, the Application Layer and the Business Layer.

Application layer

in the three-layer model



Processing

Application

Business

If this was tricky, revisit Topic 5: IoT architecture.

Q4

Self-assessment: question 4 of 5

Tests: Devices and components (Topics 4 and 6)

What is the key functional difference between a sensor and an actuator?

Pause the video

Try to answer on your own first. You retain far more by testing yourself.

about 5 seconds

5 4 3 2 1

The answer is on the next slide. No peeking!

Q4 Question 4: the answer

Tests: Devices and components (Topics 4 and 6)

What is the key functional difference between a sensor and an actuator?

A sensor converts a physical quantity into an electrical signal: it perceives the world. An actuator converts a digital command into physical action: it acts on the world.

Sensor

It perceives

Physical quantity → electrical signal

Actuator

It acts

Digital command → physical action

If this was tricky, revisit Topic 6: IoT devices explained.

Q5

Self-assessment: question 5 of 5

Tests: Data journey (Topic 7)

Name all five stages of the complete IoT data journey, in their correct order.

Pause the video

Try to answer on your own first. You retain far more by testing yourself.

about 5 seconds

5 4 3 2 1

The answer is on the next slide. No peeking!

Q5 Question 5: the answer

Tests: Data journey (Topic 7)

Name all five stages of the complete IoT data journey, in their correct order.

Data acquisition, data transmission, data storage, data analytics and real-time monitoring, typically delivered through dashboards.

1 · Acquisition → 2 · Transmission → 3 · Storage → 4 · Analytics → 5 · Monitoring

If this was tricky, revisit Topic 7: the IoT data journey.

How did you do?

Got most of them right?

If you answered most of the five without peeking at the answers, you have properly internalized Unit I.

You are ready for Unit II

Missed a few? Revisit the matching topic

Q1	**M2M versus IoT** Topic 3 · IoT versus related concepts
Q2 · Q3	**Three- and five-layer models** Topic 5 · IoT architecture
Q4	**Sensors and actuators** Topics 4 and 6 · Building blocks and devices
Q5	**The five data stages** Topic 7 · The IoT data journey

Actively testing yourself beats passively nodding along.

Preview of Unit II

IoT Communication and Networking

UNIT I

What an IoT system is made of, and how it is organized into layers.

UNIT II

How do all these separate pieces actually talk to each other, over the air?

Short-range options

Bluetooth and Zigbee.

Long-range options

LoRaWAN and cellular IoT.

Application protocols

MQTT structures the actual messages that are sent.

Wider networking

IPv6 and wireless sensor networks tie it together at scale.

Next: how physically separate devices communicate, reliably and at scale.

One-line takeaway

“Every IoT system uses the same parts, the same layers and the same five-stage data journey. Learn that skeleton and the rest follows.”

Definition | Related concepts | Building blocks | Architecture | Data journey

See you in Unit II: IoT Communication and Networking.